SHAOYI ZHENG

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EDUCATION

New York University New York, NY

Ph.D. in Computer Science, Courant Institute | Advisor: Prof. Shengjie Wang Sep 2025 -- Present

Bachelor of Science, Major in Computer Science with a Minor in Mathematics (GPA:3.9; Major GPA:4.0) Aug 2021 -- May 2025

Research Interests

World Models & Robot Learning: Learning from video world models to scale robotic experience and decision-making.
Robotic Agentic Systems: Hierarchical systems connecting long-horizon reasoning and planning with reactive control.
Efficient Foundation Models: Improving the efficiency and scalability of large generative models while preserving quality.

Selected Publications

- *De-Diffusing WAMs: Action-Sufficient Direct Prediction for Robot Control*†. Under Review at ICLR 2027.
- *Learning to Be Guided: Shaping Latent Geometry for Classifier-Free Guidance*†. Under Review at ICLR 2027.
- *ToMA: Token Merge with Attention for Diffusion Models*†. ICML 2025.
- *HilbertA: Hilbert-Curve-Aligned Sparse Attention for 2D Structured Data*†. ICML 2026.
- *InfoFlow KV: Information-Flow-Aware KV Recomputation for Long Context*†. ICML 2026.
- *Submodular Context Partitioning and Compression for In-Context Learning*†. arXiv preprint.

† First author or co-first author.

PROJECTS

De-Diffusing WAMs: Action-Sufficient Direct Prediction for Robot Control [Paper] ByteDance / NYU

First Author *Under Review at ICLR 2027*

- Investigated what predicted futures must preserve for robot control, identifying a mismatch between video-space reconstruction and the action information required by downstream policies in cascade world-action models.
- Separated stochastic sampling from iterative refinement in a cascade WAM; local sensitivity analysis showed that future-prediction errors have unequal consequences for downstream action prediction, motivating action-sufficient supervision.
- Proposed DirectWAM: distilled diffusion-pretrained video and action experts into deterministic single-step predictors, supervising student-generated futures through a frozen action model while retaining the future-to-action interface.
- Achieved 97.1% success on LIBERO and 85.4% on RoboTwin 2.0, with 28.0× lower model-side latency on one H100 (5042 to 180 ms); validated the future-to-action cascade on three physical robot manipulation tasks.

Learning to Be Guided: Shaping Latent Geometry for Classifier-Free Guidance [Paper | Web] NYU

Co-First Author *Under Review at ICLR 2027*

- Identified class geometry as a missing factor in guided diffusion representation learning: modeling the marginal latent distribution alone does not ensure that class conditions remain distinguishable under noise for effective guidance.
- Derived oracle classifier-free guidance (CFG) corrections under a Gaussian class model, relating inter-class separation and intra-class spread to noisy posteriors and a representation objective whose descent direction aligns with guidance.
- Co-developed Prototype Regularization, a pooled cosine-prototype surrogate for the guidance-aligned objective; used controlled class-center transplantation to test how latent geometry affects conditional generation under CFG.
- Reduced ImageNet 256×256 guided gFID from 3.31 to 2.28 under matched training budgets, surpassing DINOv3-aligned VA-VAE (2.44); extended the approach to pseudo-labels, video generation, and text-conditioned image synthesis.

HilbertA: Hilbert-Curve-Aligned Sparse Attention for 2D Structured Data [Paper] NYU

First Author *ICML 2026*

- Reframed sparse visual attention as a joint problem of spatial inductive bias and hardware layout: token ordering determines both sparse receptive fields and whether GPU memory reads remain contiguous and coalesced.
- Compared space-filling curves through local edge stretch and global geometric distortion, motivating Hilbert ordering as a way to map 2D spatial neighborhoods onto contiguous memory segments for efficient sparse attention.
- Developed HilbertA with local tiles, layer-wise sliding, and a shared central region as a positional anchor to retain spatial structure and enable cross-tile communication; index-shifted Triton kernels avoid repeated memory rearrangement.
- Validated quality-efficiency trade-offs across diffusion and vision-language models: up to 4.16× attention and 1.44× end-to-end speedup on Flux.1-dev (2048×2048), and 1.57× faster time to first token on Qwen3-VL-8B.

InfoFlow KV: Long-Context KV Recomputation [Paper | Code | Web] NYU / Duke / MBZUAI

Co-First Author *ICML 2026*

- Reframed selective KV recomputation as recovering cross-document information pathways: useful targets must be relevant to the query and positioned to influence generation, rather than merely differ from cached states.
- Showed that prompt-conditioned attention is an effective importance signal when scoring matches inference-time RoPE geometry; reconstructed global positions to make token rankings consistent with downstream decoding.
- Co-developed InfoFlow KV with selective recomputation and optional importance-based reordering of independent evidence chunks, then reselected tokens under the new geometry to improve their contribution to answer generation.
- Evaluated three LLM backbones and Qwen3-VL at a 15% recomputation budget; reduced 32K-context time to first token from 2350.1 to 914.0 ms (2.57× over ring attention) with sequence-parallel execution on four H100 GPUs.

ToMA: Token Merge with Attention for Diffusion Models [Paper | Code] NYU

Co-First Author *ICML 2025*

- Investigated why token reduction fails to accelerate diffusion models with optimized attention, identifying GPU-inefficient sorting and scatter operations as the gap between theoretical FLOP savings and practical latency.
- Formulated representative token selection as facility-location submodular maximization; expressed merging as an attention-based linear projection with approximate reconstruction to restore spatial resolution for iterative denoising.
- Exploited spatial coherence and repeated merge patterns to amortize selection overhead across layers and timesteps.
- Reduced end-to-end SDXL/Flux.1-dev generation latency by up to 24%/23%, with DINO score differences below 0.07.

Submodular Context Partitioning and Compression for In-Context Learning [Paper] NYU / UW / UMD

First Author *arXiv preprint*

- Investigated how random context partitioning wastes a fixed in-context learning budget through within-block redundancy and incomplete coverage across blocks, motivating structure-aware demonstration selection for efficient inference.
- Developed Sub-CP, a query-independent facility-location framework with four strategies spanning global diversity to local coherence, controlling block representation and complementarity while enabling offline precomputation.
- Evaluated DENSE, ICAE, and CEPE, finding that preferred context structure depends on the downstream framework; Local Diverse raised DENSE/PoE TREC accuracy from 38.4% to 67.6% (+29.2 percentage points).

PROFESSIONAL EXPERIENCE

ByteDance San Jose and Shanghai

Research Scientist Jun 2026 -- Sep 2026

- Conducted robotics research on video world-model representations and their use in downstream robot control.

Tencent Technology Onsite, Shanghai, China

Machine Learning Engineer Intern May 2024 -- Aug 2024

- Built multi-node diffusion pipelines with ControlNet and LoRA to generate 1M+ synthetic facial training samples.
- Fine-tuned a 1B-parameter multimodal anti-spoofing model with distributed data-parallel training on eight H100 GPUs, achieving 97% accuracy on a customized spoof-detection benchmark using the curated dataset.

SenseTime Technology Onsite, Shanghai, China

Machine Learning Engineer Intern May 2023 -- Aug 2023

- Curated an 8M text QA corpus for customer service chatbots using contextual augmentation and embedding-based filtering.
- Fine-tuned a chatbot on the curated corpus, improving downstream F1 score by 7% in production evaluation.

SKILLS

Distributed Training: Multi-GPU/multi-node training, memory optimization, checkpointing, and distributed debugging.

Inference Optimization: vLLM; CUDA/Triton kernels, performance profiling, and latency/memory optimization.

Robotics Systems & Data Collection: Simulation debugging in LIBERO and RoboTwin 2.0; real-robot debugging on Galaxea A1, including teleoperation and leader-follower demonstration data collection for manipulation.